ColdChain — Vaccine & Medicine Storage Monitor

Total Marks: 100

Problem Statement

Vaccines and insulin can lose effectiveness when stored outside **2–8°C**. Build an **IoT cold-chain system** that monitors fridge conditions, tracks vaccine heat exposure, maintains safe temperature automatically, and provides a complete web application.

Tasks

1. Fridge Node

- Use ESP32 for two fridges with temperature sensor, door switch, compressor relay, and LED.
- Display temperature and **Safe / Warning / Breach** on OLED.
- Send MQTT data every 5 seconds and on door-state changes.
- Compressor: **ON at 7°C, OFF at 3°C**, with a 60-second minimum off-time.

2. Heat-Exposure Budget

- Each batch has a **60-minute budget above 8°C**.
- Track exposure and remaining budget.
- Exposure above **12°C counts double**.
- Below 20%: **At Risk**. At 0%: **Discard** and keep the red LED ON until admin reset.

3. Power Failure & Offline Mode

- Simulate power failure with a button and report battery/power status.
- Buffer at least **30 minutes of readings** during Wi-Fi loss and sync them in order without duplicates.
- Door open for over **60 seconds** → buzzer + SMS, Telegram, or email alert.

4. ColdChain Web Application

- **Auth & Roles:** JWT + bcrypt with Admin, Pharmacist, and Inspector roles. Enforce permissions on the server.
- **Live Dashboard:** Temperature, door, power, compressor, and heat-budget status. And a well designed UI.
- **Inventory:** Manage vaccine batches and assign or move them between fridges.

- **Public QR Check:** Show batch **Safe / At Risk / Discard** status and expiry without login.
- **Device Security:** HMAC-SHA256 + timestamp + nonce; reject and log invalid or replayed messages.
- **Tamper-Evident Logs:** Use a hash chain and provide **Verify Integrity** to detect edited records.

5. AI Breach Prediction & Relocation

- Predict the fridge temperature **30 minutes ahead** and give a warning if it may go above 8°C.
- Detect if the compressor is becoming weaker and show a **0–100 health score**.
- If a fridge may become unsafe, suggest **which batches to move and where to move them**.
- Pharmacist can confirm the suggested relocation with **one click**.
- Provide a short report showing the **dataset, features, and results**.